

Midterm review
COSC 612/AIT 624
Fall 2026

Chapter 1 – Introduction of software engineering

Expected Questions – short answers.

1. What is software
2. The 4 P's of software engineering
3. Software change and evolution
4. Software Engineering Ethics

Chapter 2 – Software Process

Expected Questions – Case study, MCQs, short answers

1. Understand the activities involved in software engineering.
2. Software Process Description
3. Plan driven vs. Agile.
4. Process model – Waterfall, Incremental, Re-use Oriented.
5. Coping with change – costs of reworks

Chapter 3 – Agile Methodology

Expected Questions – MCQs, short answers.

1. Agile Principles
2. Extreme Programming and its practices
3. Scrum – components of scrum methodology.

Chapter 4 – Requirements Engineering

Expected Questions – Case study, short answers

1. *Different types of requirements – user and system requirements, functional and non-functional requirements
2. Requirements Elicitation – interviewing, issues with stakeholders and Natural language specification
3. Requirements Specification – User case template and diagrams
4. Requirements validation – Testing requirements
5. Requirements management planning

Chapter 5– Architecture Design

Expected questions: MCQs, short answers, T/F

1. System views
2. Application architecture
3. Functional vs. Object Oriented design

Chapter 6 – System Modeling

Expected Questions – Case Study and MCQs

1. System perspectives and Context models.
2. Use Case diagrams.